

Canary Current Base model validation

Field	Entry
Region	Canary Current (LME 027) Regional workbook
Catch source	Sea Around Us retained regional dataset, 1950–2019; landings reference year 2019. Provider revision unspecified. Catch archive Catch and coefficient context
Selected article	Candidate article: Guénette S, Meissa B, Gascuel D (2014). Assessing the Contribution of Marine Protected Areas to the Trophic Functioning of Ecosystems: A Model for the Banc d'Arguin and the Mauritanian Shelf. PLoS ONE 9(4), e94742. DOI Article PDF Supplement
Other known articles	Morissette, Melgo, Kaschner, Gerber & Bamy (2009): Northwest African food-web model; the currently selected EcoBase 118 version represents 1987. Villanueva, Tito-de-Morais, Weigel & Moreau (2004): An Ecopath model of the Sine-Saloum Delta biosphere reserve (Senegal). Central article records Retained source folders
Selected model	Candidate: Guenette2014_BancArguin_Base_1991. 51 source groups, 47 consumers and 3 reported fleets; Mauritanian shelf to 200 m, approximately 33,224 km ² . Baseline 1991; Ecosim calibration 1991–2006. Canonical model JSON Source extraction
Other models from the same article	M30 lowers assumed subtidal benthos density by 30%; P30 raises it by 30%. Both change Banc feeding proportions and selected balanced biomasses. All three share the 1991 shelf domain. EcoBase 689 is a separately deposited numerical version, with material differences from the printed tables.
Selection rationale	Article – ? Candidate review requested; no recorded decision selects this article. Model – ? No decision selects this variant. Overview retains Northwest Africa 1987 after the earlier comparison recorded failures for the Banc d'Arguin configurations. Current selection record Exact recorded rationale
Model extraction	Table 1 and S8 Base conflict; for BA phytoplankton, EE is 0.260 versus 0.40. Both values and their table contexts remain documented. S2 prints Groupers ad → BA L crustaceans as 1.34 (diet sum 2.2006); Coastal birds sum to 0.980331. Four blank cells remain unknown. The source matrix was not repaired. Multi-stanza Z is distinct from P/B; missing P/B and growth/transition details are retained separately. No discard observations were reported; unknowns remain distinct from zero. Source conflicts Diet cell evidence Online recovery

Field	Entry
GE diagnostics	Source input: NOT_RUN. Five published diet rows fail the admission check. Separate runtime experiment: FAIL. No negative SPPR entries across the basal-source columns; all 52×5 entries are finite. $\rho_{\text{living}} = 0.5567$ $b = 0.4356$ Detritus (51): SPPR = 2.2633. Strict model-input balance fails; numerical convergence does not validate the source. Source admission Runtime experiment
TE diagnostics	Source input: NOT_RUN. Five published diet rows fail the admission check. Separate runtime experiment: FAIL. No negative SPPR entries across the basal-source columns; all 52×5 entries are finite. $\rho_{\text{living}} = 0.8002$ Detritus (51): SPPR = 0.5504. Strict model-input balance fails; numerical convergence does not validate the source. Source admission Runtime experiment
SPPR calculation	[Researcher calculation choices and notes]
Geographic fit	A. Region covered by study: approximately 3.0% (about 2.5–3.5%). B. Study area covered by region: approximately 95–100%.
Temporal fit	The baseline represents 1991 and calibration spans 1991–2006; the mapping reference is 2019. Transferring fixed coefficients and source catch/biomass proportions across years and across the much larger LME is an explicit approximation.
Other	The experiment normalizes a runtime diet copy and approximates missing P/B as $Q/B \times P/Q$; printed P/B = 0 remains unchanged. It is not the published Base parameterization. With Egestion also FAILS. EcoBase 689 also fails the flat-model checks and does not establish source validation. Diagnostic-FAIL experiment, 2019 landings, all sources: GE PPR 105.84 million t C, TE 1,362.76 million t C and With Egestion 55.02 million t C; respectively 22%, 290% and 12% of ensemble NPP (472.96 million t C). No validated source-model estimate is available. The currently selected Northwest Africa 1987 model was disqualified by Ido Carmel on 02/10/2026: “Reason – too strong living compartments recycling due to near-zero EE values ($\rho_{\text{living}} = 0.99$).” This is a separate model’s review. Regional assessment Independent native-deposit audit
Open issues and next action	
Review and reproducibility	Researcher name: [name] review date: [dd/mm/yyyy]

Taxon mapping and coverage

Reference: 2019 landings; all retained regional catch labels, including zero landings.

51 source groups; 512 catch taxa. Computational import groups are separate.

Simple-chain PPR totals 244,709,206 t C in 2019.

100 labels lack a saved TL/coefficient. All have zero recorded 2019 landings and contribute zero annual PPR; the annual total is complete.

Method: independent simple trophic chain, using retained taxon coefficients.

[Excel taxon appendix and descriptive Sources sheet](#)

Overall confidence	Taxa (n)	Catch (%)	Simple-chain PPR (%)
High	116	49%	37%
Medium	283	25%	39%
Low	45	1.7%	9.8%
Very low	68	24%	14%
Unresolved	0	0%	0%

Overall confidence uses the weaker required membership or allocation component.

Group assignment rules

Plain-language rule	Confidence	PPR percentage
Explicit species assignment in the source composition table	High	37%
Supported extension from listed regional representatives	Medium	20%
Supported ecological extension to a documented guild	Medium	14%
Broad reporting category approximated across compatible groups	Very low	11%
Partial or conflicting ecological fit	Low	9.8%
Assumed eligible group set supported by source taxonomy and ecology	Medium	3.1%
Unambiguous fit to the documented residual cephalopod group	High	2.6%
Closest represented analogue despite habitat or taxonomic mismatch	Very low	2.1%
Verified synonym of an explicitly listed source species	High	0.20%

Membership evidence: source Tables S1, S4 and S5, documented guild definitions and regional taxonomic evidence. [Source supplement](#)

Allocation weight rules

Plain-language rule	Confidence	PPR percentage
One eligible group; no allocation split needed	High	83%

Plain-language rule	Confidence	PPR percentage
Source model 1991 reported catch proportions, assumed across target years and bases	Medium	17%
Model-biomass proportions when complete candidate catches have zero total	Medium	0.047%

Weights and assumptions: source 1991 catch proportions; complete model biomass where candidate catch totals are zero. The spatial, temporal and life-stage transfer is assumed. Juvenile source zeros do not show that later catches or discards contain no juveniles. [Allocation evidence](#)

Each percentage is the independent simple-chain PPR associated with taxa using that rule.

No unresolved mappings remain. The 68 Very low assignments account for 24% of catch and 14% of simple-chain PPR.

Very low decisions

Affected taxa	Why confidence is very low
Actinopterygii	Broad ray-finned-fish reporting spans named fish, pelagic, coastal and shelf pools. Approximate with source fish pools 3–29 except 11/20 (selacians), which are taxonomically excluded. Constituent composition is unobserved.
Alepisaurus ferox; Brama brama; Bramidae; Lampris guttatus	Bathypelagic or oceanic midwater fish lack their own depth/feeding compartment. Pelagic L is the closest mobile pelagic-fish analogue, with deep-water/feeding differences retained.
Alopias; Alopias superciliosus; Alopias vulpinus; Carcharhinus falciformis; Carcharhinus galapagensis; Carcharhinus longimanus; Carcharhinus signatus; Carcharodon carcharias; Cetorhinus maximus; Isurus; Isurus oxyrinchus; Isurus paucus; Lamna nasus; Lamnidae; Lamniformes; Mobula birostris; Odontaspis ferox; Prionace glauca	Oceanic, highly migratory or large filter-feeding elasmobranch lacks a matching pool. Shelf selacians is the closest offshore cartilaginous-fish analogue; oceanic/depth or feeding differences (especially basking shark/mobula) are material. Coastal selacians would not resolve those differences.
Antimora rostrata; Mora moro; Trachyrincus scabrus	Deep slope fish exceed the modeled shelf domain. Shelf L includes a deep-associated macrourid and provides a meaningful predatory demersal-fish analogue, with depth/composition mismatch retained.
Aristaeomorpha foliacea; Aristaeopsis edwardsiana; Aristeidae; Aristeus antennatus; Aristeus varidens; Chaceon affinis; Chaceon maritae; Geryon; Paromola cuvieri	Deep/slope large crustaceans extend beyond the modeled <200 m shelf and lack a deep-water pool; shelf large crustaceans are the closest represented benthic decapod analogue. This habitat mismatch is retained.
Carcharhinidae; Carcharhiniformes; Carcharhinus; Squaliformes	Broad shark reporting spans source coastal/shelf habitats and potentially oceanic/deep species. Both selacian pools are a meaningful approximation; no observed species/habitat composition supports stronger containment.

Affected taxa	Why confidence is very low
Centrophorus; Centrophorus granulosus; Centrophorus lusitanicus; Centrophorus squamosus; Centroscyllium fabricii; Centroscymnus coelolepis; Centroselachus crepidater; Dalatias licha; Deania calceus; Echinorhinus brucus; Etmopterus; Etmopterus princeps; Scymnodon ringens; Somniosus rostratus	Deep-water dogfish/shark assemblage extends beyond the modeled shelf. Shelf selacians (including Squalus/Oxynotus/Galeus) is the closest taxonomic/demersal analogue, with depth and unlisted-species mismatch.
Chimaeriformes	Chimaeras have no source group and differ taxonomically from selacians; shelf selacians are the closest cartilaginous demersal-fish analogue, with deep-water and feeding mismatch.
Chondrichthyes	Broad cartilaginous-fish label includes sharks/rays in both source pools and potentially unrepresented chimaeras. Approximate with both meaningful selacian pools; disclose unrepresented component.
Decapoda	Broad decapod reporting can include large harvested shrimps/crabs/lobsters and smaller benthic forms. Use large and small Banc/shelf crustacean pools; cannot infer strict large-only containment from provider commercial tag. Exclude pelagic macrozooplankton because source macro examples are euphausiids/mysids rather than harvested adult Decapoda.
Lepas	Pelagic/rafting goose barnacles are crustaceans without a matching sessile-epifaunal compartment. Small benthic crustacean pools are the closest size/taxonomic analogue, with habitat and feeding mismatch.
Malacostraca; Miscellaneous marine crustaceans	Composition unspecified across large and small benthic crustaceans and macrozooplankton (euphausiids/mysids). Retain those compatible compartments; source catch labels do not establish a complete size/habitat boundary. Meio/larval pools excluded as not a plausible adult fishery reporting population.
Marine finfishes not identified; Marine fishes not identified	Unidentified finfish can span named fish, pelagic, coastal and shelf pools. No observed composition supports a narrower exclusion; approximate with all source fish pools (3–29), including selacians.
Marine groundfishes not identified	Groundfish reporting establishes demersal tendency but no species mix. Include compatible named and generic coastal/shelf fish groups, including demersal elasmobranchs; exclude strictly pelagic/mullet pools.
Marine pelagic fishes not identified	Pelagic reporting spans migratory mullets, the five named pelagic pools, Coastal M (Ethmalosa), Shelf M (explicit pelagic/benthopelagic jacks) and offshore selacian analogue (potential pelagic sharks/rays). Pool names do not exclude these known exceptions. Unobserved composition is a broad approximation; strictly benthic fish pools are excluded.

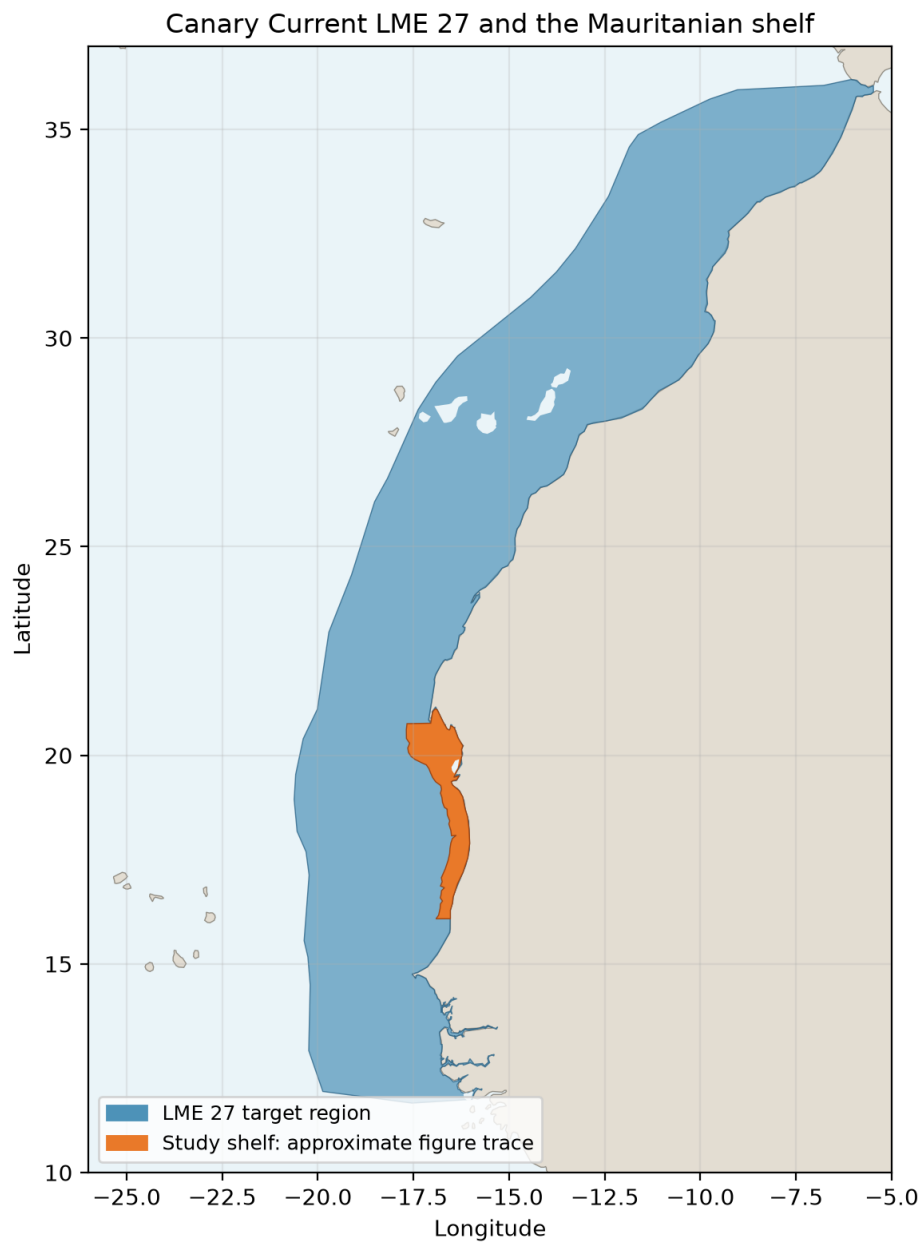
Affected taxa	Why confidence is very low
Miscellaneous aquatic invertebrates	Unidentified invertebrate reporting can include cephalopods, large/small crustaceans, molluscs, worms, echinoderms and other macro-invertebrates. The provider tag is not exhaustive; named and residual compatible macrofaunal pools retained. Meio/mesoplankton and primary producers excluded as unsupported harvested reporting components.
Mollusca	The taxonomic label can include cephalopods and benthic molluscs. Provider demersal-invertebrate tag does not prove cephalopods absent. Retain both named cephalopod groups and Banc/shelf mollusc pools in a broad composition approximation.
Myctophidae	Mesopelagic lanternfish are not a represented source fish pool. Sardinelles/anchovy supplies the closest small schooling zooplanktivorous-fish analogue, despite depth and diel-migration mismatch.
Octopoda	Broad order includes benthic octopuses represented by <i>Octopus vulgaris</i> and potentially pelagic argonaut/octopod forms in other cephalopods; unobserved composition is approximated across both.
Perciformes	Historical broad perch-like reporting spans many source fish pools. Approximate using compatible named/guild compartments, excluding clupeiforms, selacians, catfish and flatfish-only groups; no observed constituent mix is available.

Geographic evidence

The study covers a narrow Mauritanian shelf portion of the Canary Current LME. Approximate coverage accounts for land exclusion and the shelf's 200 m offshore boundary. [Boundary and area evidence](#)

Region area

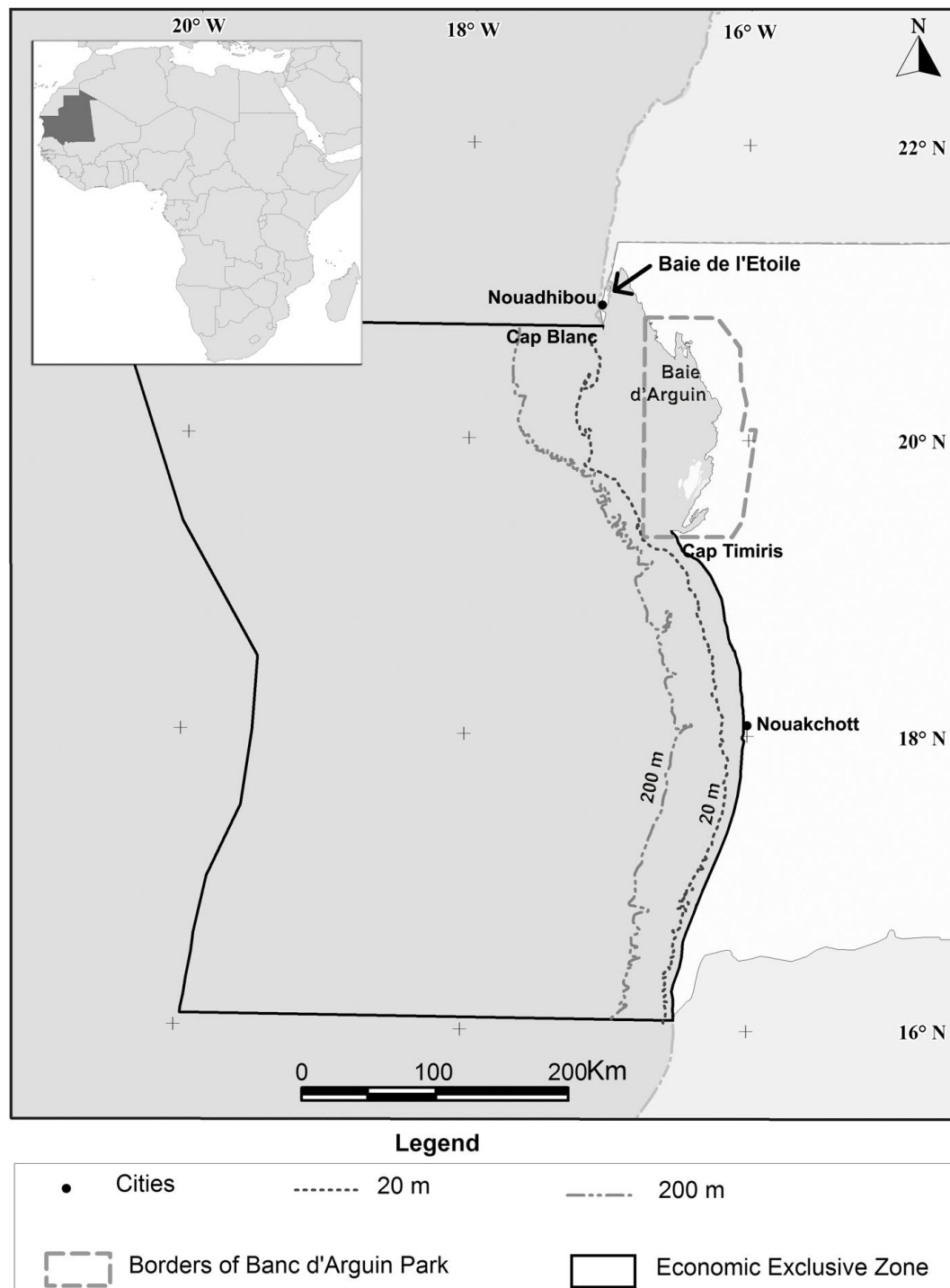
Target region boundary



Canary Current LME 27 in blue; approximate study shelf in orange. Project LME boundary and Natural Earth 1:50m land mask. [Regional comparison](#)

Study area in the article

Study boundary for the selected model



Guénette, Meissa & Gascuel (2014), Figure 1, PDF page 3. The study area is the shelf shallower than 200 m, approximately 33,224 km² including the 6,450 km² Banc. The solid black offshore line marks the larger EEZ.
[Original figure](#)